# Time Series Project — Phase 3

## Task 1 — Data Exploration

| Property | Value |
|---|---|
| Total number of clips | 8732 |
| Number of classes | 10 |
| Sampling rate (Hz) | 22050 |
| Min clip duration (s) | 0.055 |
| Max clip duration (s) | 4.000 |
| Mean clip duration (s) | 3.608 |

**Note:** Class distribution is imbalanced — gun_shot (~370) and car_horn (~430) are under-represented vs. other classes (~1000).

## Task 2 — Spectrogram Generation

**Observation:** STFT concentrates detail in low frequencies; Mel spreads detail uniformly. Transients show vertical bursts, sustained sounds show horizontal bands.

## Task 3 — Transfer Learning Results (n_fft = 1024)

| Spectrogram Type | Train Accuracy | Val Accuracy | Test Accuracy | Verdict |
|---|---|---|---|---|
| STFT | 84.84% | 72.79% | 72.28% | |
| Mel | 88.29% | 74.39% | 76.70% | **Mel ✓** |

**Comment:** Mel outperforms STFT by ~4 percentage points on test accuracy.

**Confusion matrix:** see notebook Cell 30 (STFT and Mel test-set confusion matrices).

## Task 4 — Window Size Experiment

| Window Size (n_fft) | STFT Train | STFT Test | Mel Train | Mel Test |
|---|---|---|---|---|
| 256 | 85.00% | 74.07% | 81.75% | 73.95% |
| 512 | 83.57% | 72.64% | 82.82% | 71.92% |
| 1024 (default) | 86.35% | 75.03% | 85.87% | 77.54% |
| 2048 | 80.18% | 68.94% | 88.26% | 76.70% |

**Q1 — Shape effect:** Smaller n_fft → more time frames, fewer frequency bins. Larger n_fft → the opposite (time–frequency uncertainty principle).

**Q2 — Best window per type:** STFT → n_fft = 1024 (75.03%). Mel → n_fft = 1024 (77.54%).

**Q3 — Consistent trend:** Yes. Both peak at n_fft = 1024, and Mel matches or beats STFT at every window size.

## Project Result

In Phase 3, I implemented a transfer learning pipeline using a frozen ImageNet-pretrained EfficientNet-B0 with only the final 10-class head trained on UrbanSound8K spectrograms. I compared STFT and Mel representations across four window sizes (256, 512, 1024, 2048). Mel matched or outperformed STFT at every window size tested, and both spectrogram types peaked at n_fft = 1024 — STFT at 75.03% and Mel at 77.54%. The overall best configuration was Mel at n_fft = 1024, confirming the perceptual advantage of mel-scaled features for pretrained image backbones: spreading useful information uniformly across the image transfers better than concentrating it in a thin low-frequency band.